Mathematics HW #1

A(1)

$$a_{ij} = i - j,$$

$$b_{ij} = \frac{i}{j}$$

$(a_{ij} \in A,\ b_{ij} \in B,\ (i,j = 1, 2, 3))$

$$A = \begin{pmatrix} 0 & -1 & -2 \\ 1 & 0 & -1 \\ 2 & 1 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ 2 & 1 & \frac{2}{3} \\ 3 & \frac{3}{2} & 1 \end{pmatrix}$$

(2) A+B, AB, BA

$$A+B = \begin{pmatrix} 1 & -\frac{1}{2} & -\frac{5}{3} \\ 3 & 1 & -\frac{1}{3} \\ 5 & \frac{5}{2} & 1 \end{pmatrix}$$

$$AB = \begin{pmatrix} 0 & -1 & -2 \\ 1 & 0 & -1 \\ 2 & 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ 2 & 1 & \frac{2}{3} \\ 3 & \frac{3}{2} & 1 \end{pmatrix} = \begin{pmatrix} -8 & -4 & -\frac{8}{3} \\ -2 & -1 & -\frac{2}{3} \\ 4 & 2 & \frac{4}{3} \end{pmatrix}$$

$$BA = \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ 2 & 1 & \frac{2}{3} \\ 3 & \frac{3}{2} & 1 \end{pmatrix}\begin{pmatrix} 0 & -1 & -2 \\ 1 & 0 & -1 \\ 2 & 1 & 0 \end{pmatrix} = \begin{pmatrix} \frac{7}{6} & -\frac{2}{3} & -\frac{5}{2} \\ \frac{7}{3} & -\frac{4}{3} & -5 \\ \frac{7}{2} & -2 & -\frac{15}{2} \end{pmatrix}$$

(3) $\det A = (-1)^{1+1} \cdot 0 \cdot (0-(-1)) + (-1)^{1+2} \cdot (-1) \cdot (0-(-2))$

$$+ (-1)^{1+3} \cdot (-2) \cdot (1-0)$$

$$= 2 - 2 = 0$$

B-1

$$A = \begin{pmatrix} 1 & 3 & 2 \\ 2 & 2 & 0 \\ -3 & 1 & 1 \end{pmatrix}$$

$C_{ij} = (-1)^{i+j} M_{ij}$

$\text{adj}(A) = C^T$

$A^{-1} = \dfrac{\text{adj}(A)}{\det(A)}$

① $\det A = (-1)^{1+3} \cdot 2 \cdot (2-(-6)) + (-1)^{(2+3)} \cdot 0 \cdot (1-(-9)) + (-1)^{3+3} \cdot 1 \cdot (2-6)$

$= 16 + 0 - 4$

$= 12$ //

② $C = \begin{pmatrix} 2 & -2 & 8 \\ -1 & 7 & -10 \\ -4 & 4 & -4 \end{pmatrix}$

$\downarrow\downarrow$

$$\text{adj } A = \begin{pmatrix} 2 & -1 & -4 \\ -2 & 7 & 4 \\ 8 & -10 & -4 \end{pmatrix}$$

$$\therefore A^{-1} = \frac{1}{12}\begin{pmatrix} 2 & -1 & -4 \\ -2 & 7 & 4 \\ 8 & -10 & -4 \end{pmatrix} = \begin{pmatrix} 1/6 & -1/12 & -1/3 \\ -1/6 & 7/12 & 1/3 \\ 2/3 & -5/6 & -1/3 \end{pmatrix}$$

B-2 (REF)

$x_1 + 3x_2 + 2x_3 = 4$

$-2x_1 + 3x_2 + 6x_3 = -11$

$3x_1 + x_2 + x_3 = 0$

$$\left(\begin{array}{ccc|c} 1 & 3 & 2 & 4 \\ -2 & 3 & 6 & -11 \\ 3 & 1 & 1 & 0 \end{array}\right)$$

$R_2 = 2R_1 + R_2$

$R_3 = -3R_1 + R_3$ $\rightarrow$

$$\left(\begin{array}{ccc|c} 1 & 3 & 2 & 4 \\ 0 & 9 & 10 & -3 \\ 0 & -8 & -5 & -12 \end{array}\right)$$

$R_3 = +\frac{8}{9}R_2 + R_3$

$$\left(\begin{array}{ccc|c} 1 & 3 & 2 & 4 \\ 0 & 9 & 10 & -3 \\ 0 & 0 & \frac{35}{9} & -\frac{44}{3} \end{array}\right)$$

$\therefore x_1 + 3x_2 + 2x_3 = 4$

$9x_2 + 10x_3 = -3$

$\frac{35}{9}x_3 = -\frac{44}{3}$

$\therefore \quad x_1 = -\frac{1}{35}$,

$x_2 = \frac{211}{11}$,

$x_3 = -\frac{132}{35}$ //

B-3 (RREF)

B-2 에서 !

$$\left(\begin{array}{ccc|c} 1 & 3 & 2 & 1 \\ 0 & 9 & 10 & -3 \\ 0 & 0 & \frac{35}{9} & -\frac{44}{3} \end{array}\right) \xrightarrow{R_2 = \frac{1}{9}R_2} \left(\begin{array}{ccc|c} 1 & 3 & 2 & 1 \\ 0 & 1 & \frac{10}{9} & -\frac{1}{3} \\ 0 & 0 & \frac{35}{9} & -\frac{44}{3} \end{array}\right)$$

$R_1 = -3R_2 + R_1$

$R_3 = \frac{9}{35}R_3$

$$\left(\begin{array}{ccc|c} 1 & 0 & -\frac{4}{3} & 5 \\ 0 & 1 & \frac{10}{9} & -\frac{1}{3} \\ 0 & 0 & 1 & -\frac{132}{35} \end{array}\right) \xrightarrow[R_2 = -\frac{10}{9}R_3 + R_2]{R_1 = \frac{4}{3}R_3 + R_1} \left(\begin{array}{ccc|c} 1 & 0 & 0 & -\frac{1}{35} \\ 0 & 1 & 0 & \frac{211}{11} \\ 0 & 0 & 1 & -\frac{132}{35} \end{array}\right)$$

$\therefore \quad x_1 = -\frac{1}{35}$

$x_2 = \frac{211}{11}$

$x_3 = -\frac{132}{35}$ //

B-4

$H = \{(a,b,0) \in R^3 \mid a, b \in \mathbb{R}\}$

① 0 벡터 포함? → $(0,0,0) \in H$ → O

② 덧셈에 닫힘? → $(a_1,b_1,0) + (a_2,b_2,0) = (a_1+a_2, b_1+b_2, 0) \in H$ → O

③ 스칼라 곱 닫힘? → $c(a,b,0) = (ca, cb, 0) \in H$ → O

$\therefore$ H는 R^3의 부분공간이다.

5.

$x_1 = (1,2,3)$, $x_2 = (-1,0,2)$, $x_3 = (0,2,4)$

$c_1x_1 + c_2x_2 + c_3x_3 = 0$ 을 만족하는 c_1, c_2, c_3 존재 → 선형종속

존재 X → 선형독립

$$c_1(1,2,3) + c_2(-1,0,2) + c_3(0,2,4) = 0$$

$$c_1 - c_2 = 0 \quad \cdots ①$$

$$2c_1 + 2c_3 = 0 \quad \cdots ②$$

$$3c_1 + 2c_2 + 4c_3 = 0 \quad \cdots ③$$

①, ②로부터 $c_1 = c_2 = -c_3$ //

③에 대입 $3c_1 + 2c_1 - 4c_1 = 0$

$\therefore c_1 = 0$

$c_2 = 0$

$c_3 = 0$

$\therefore$ $c_1 = c_2 = c_3 = 0$ 인 해만 존재 → 선형독립 //

B-6

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 2x_1 - 3x_2 \\ -x_1 + x_2 \end{bmatrix}$$

(1) $T(x) = Ax$

$$T\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}2\\-1\end{pmatrix} \qquad T\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}-3\\1\end{pmatrix}$$

$$x = \begin{pmatrix}x_1\\x_2\end{pmatrix} = x_1\begin{pmatrix}1\\0\end{pmatrix} + x_2\begin{pmatrix}0\\1\end{pmatrix}$$

$$T(x) = T\left(x_1\begin{pmatrix}1\\0\end{pmatrix} + x_2\begin{pmatrix}0\\1\end{pmatrix}\right) = x_1T\begin{pmatrix}1\\0\end{pmatrix} + x_2T\begin{pmatrix}0\\1\end{pmatrix} = x_1\begin{pmatrix}2\\-1\end{pmatrix} + x_2\begin{pmatrix}-3\\1\end{pmatrix}$$

$$\therefore T(x) = \begin{pmatrix}2 & -3\\-1 & 1\end{pmatrix}\begin{pmatrix}x_1\\x_2\end{pmatrix} \text{ //}$$

$T\begin{pmatrix}0\\0\end{pmatrix}$, $T\begin{pmatrix}2\\1\end{pmatrix}$

$$T\begin{pmatrix}0\\0\end{pmatrix} = \underset{2\times2}{\begin{pmatrix}2 & -3\\ -1 & 1\end{pmatrix}}\underset{2\times1}{\begin{pmatrix}0\\0\end{pmatrix}} = \begin{pmatrix}0\\0\end{pmatrix}$$

$$T\begin{pmatrix}2\\1\end{pmatrix} = \begin{pmatrix}2 & -3\\ -1 & 1\end{pmatrix}\begin{pmatrix}2\\1\end{pmatrix} = \begin{pmatrix}1\\-1\end{pmatrix}$$

B-7

$$A = \begin{pmatrix}1 & 1\\ 0 & 1\end{pmatrix}$$

(1) 행렬 A를 기반한 선형변환 결과 기술

$$i_t = \begin{pmatrix}1\\0\end{pmatrix},\quad j_t = \begin{pmatrix}1\\1\end{pmatrix}$$

$$\begin{pmatrix}x_1\\x_2\end{pmatrix} = x_1\begin{pmatrix}1\\0\end{pmatrix} + x_2\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}x_1+x_2\\x_2\end{pmatrix}$$

$$\therefore A\vec{x} = \begin{pmatrix}1 & 1\\ 0 & 1\end{pmatrix}\begin{pmatrix}x_1\\x_2\end{pmatrix} = \begin{pmatrix}x_1+x_2\\x_2\end{pmatrix} \text{ //}$$

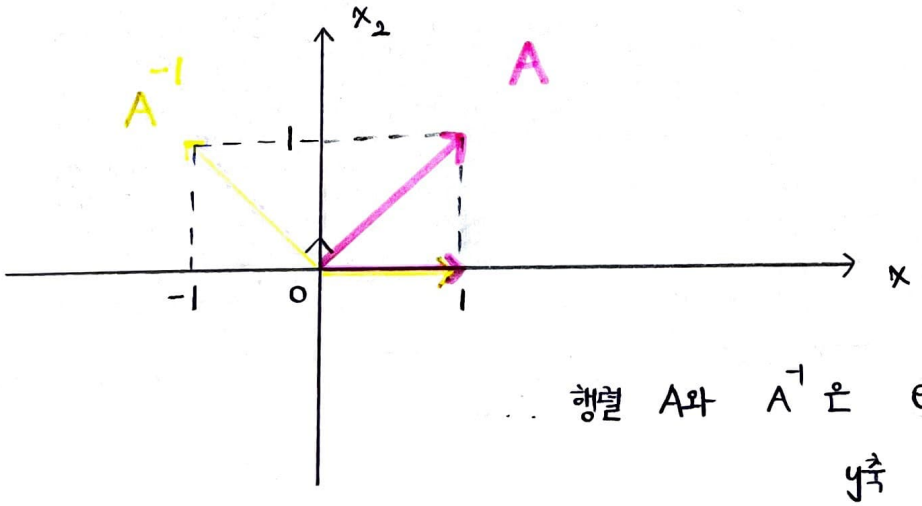
x_1 성분 x_1+x_2로 이동 (평행이동)

x_2 성분 x_2로 유지

(2) A^{-1} ?

$$A^{-1} = \frac{1}{ad-bc}\begin{pmatrix}d & -b\\ -c & a\end{pmatrix} = \frac{1}{1-0}\begin{pmatrix}1 & -1\\ 0 & 1\end{pmatrix} = \begin{pmatrix}1 & -1\\ 0 & 1\end{pmatrix} \text{ //}$$

(3)



∴ 행렬 A와 A^{-1}은 e_j 성분에 대하여 y축 대칭이다.

C-1. $\text{proj}_y x$?

$x = (1,2,3)$, $y = (2,-2,1)$

$$\text{proj}_y x = \frac{x \cdot y}{y \cdot y} y$$

$$= \frac{(1\cdot 2 + 2\cdot(-2) + 3\cdot 1)}{\sqrt{4+4+1}} (2,-2,1)$$

$$= \frac{1}{3}(2,-2,1)$$

$$= (2/3, -2/3, 1/3) \text{ //}$$

C-2.

$(x^2-1, 2)$ 가 $(2, x-1)$과 (x, x^2+3x-4) 와 직교, $x = ?$

① $(x^2-1, 2) \cdot (2, x-1) = 0$

$$\begin{cases} 2(x^2-1) = 0 & \cdots \ x = 1, -1 \\ 2(x-1) = 0 & \cdots \ x = 1 \end{cases}$$

② $(x^2-1, 2) \cdot (x, x^2+3x-4) = 0$

$$\begin{cases} x(x^2-1) = 0 & \cdots \ x = 0, 1, -1 \\ 2(x^2+3x-4) = 0 & \cdots \ x = 1, -4 \end{cases}$$

공통인 해는 1이므로 $x = 1$ //

3.

$\|x\| = 4, \quad \|y\| = 1, \quad \theta = \frac{\pi}{4}$

$(2x+3y) \cdot (x-2y) = ?$

$(2x+3y) \cdot (x-2y)$

$= 2\|x\|^2 - 6\|y\|^2 - 1 \cdot x \cdot y$

$= 2 \cdot 16 - 6 \cdot 1 - \|x\|\|y\| \cos\frac{\pi}{4}$

$= 32 - 6 - 4 \cdot \frac{\sqrt{2}}{2}$

$= 26 - 2\sqrt{2}$ //

C-4

A → QR 분해

직교 행렬 Q와 상삼각 행렬 R로 분해

$$A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

① $\vec{a_1}$ 정규화

$\vec{a_1} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \qquad \vec{a_2} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$\|a_1\| = \sqrt{2} \Rightarrow \vec{q_1} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

② $\vec{a_2}$를 $\vec{q_1}$에 대해 직교화

$\text{proj}_{\vec{q_1}} \vec{a_2} = (\vec{a_2} \cdot \vec{q_1})\vec{q_1} = 0 \qquad \therefore \vec{q_2} = \frac{a_2}{\|a_2\|} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$\therefore Q = (\vec{q_1}, \vec{q_2}) = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}$

$R = Q^T A$

$Q^T = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}$

$\therefore R = Q^T A = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} \sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix}$